

Quarterly Performance Report

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1 Overview

This report summarises system performance metrics for Q1 2026. All measurements were taken under production load conditions.

2 Latency

Average API response time improved significantly after the caching layer was introduced:

$$\bar{t}_{\text{after}} = 42 \text{ ms} \quad \text{vs} \quad \bar{t}_{\text{before}} = 310 \text{ ms}$$

This represents an **86% reduction** in mean latency.

3 Results

Endpoint	P50 (ms)	P95 (ms)	P99 (ms)
/api/compile	42	120	340
/api/status	3	8	15
/api/health	1	2	4

Table 1: API latency percentiles after optimisation.

4 Error Rate

Total error rate dropped from 2.3% to 0.04% across all endpoints.

5 Conclusion

The caching and connection pooling improvements exceeded targets. Next quarter will focus on reducing P99 latency below 200 ms.

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